

KNOWLEDGE ENGINEERING AND ONTOLOGIES COURSE
PROJECT REPORT

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Executive Summary (2 pts)

This project presents a semantic solution for managing academic conference data through a custom-built OWL ontology and an interconnected knowledge graph. The system is engineered to structure complex, heterogeneous relationships between academic papers, authors, sessions, and research topics. By utilizing the OntoClean methodology and METHONTOLOGY principles, the project ensures a logically consistent TBox and a data-rich ABox that models dynamic user roles without logical conflicts. Key features include automated data acquisition strategies from external scholarly APIs, complex SPARQL querying for schedule management and conflict-of-interest detection, and rigorous data quality enforcement through SHACL validation shapes. Furthermore, an innovative integration with Large Language Models (LLMs) via a Retrieval-Augmented Generation (RAG) workflow bridges the gap between unstructured PDF documents and structured knowledge graphs, utilizing closed-loop validation to mitigate hallucinations. The final outcome provides a highly scalable and reproducible semantic platform implemented in GraphDB, allowing academic organizations to enhance information retrieval, automate administrative constraints, and significantly improve data-driven decision-making processes in conference management. Ultimately, this engineering work establishes how semantic web

technologies can minimize operational overhead while guaranteeing strict logical consistency in highly dynamic domain environments.

Description of the project (2,5 pts)

The primary objective of this engineering project is to design, implement, and validate a highly sophisticated, semantically enriched Knowledge Graph tailored for academic conference management. Managing modern academic conferences involves coordinating a vast array of heterogeneous and deeply interconnected data, including research paper submissions, peer-review pipelines, author profiles, multi-track schedules, session allocations, and diverse scientific domains. Traditional relational databases struggle with these environments due to rigid schemas, data silo creation, and an inherent lack of semantic reasoning capabilities, which are essential for automatically detecting conflicts of interest or discovering implicit track categorizations.

To address these limitations, our system introduces a formal semantic web framework built using Protégé for comprehensive TBox development and executed via GraphDB as a high-performance triplestore for robust ABox management. The scope of the project encompasses an end-to-end knowledge engineering pipeline consisting of four major components: Data Acquisition Layer, Semantic Reasoning Infrastructure, Data Quality/Validation Layer, and Innovative LLM Integration.

2.1. Changes and Improvements in Version 2 (v2)

Compared to the initial phase (v1) of the Academic Conference Management Ontology, the following advanced enhancements have been successfully implemented to build a comprehensive, production-ready semantic system for Version 2:

- **Schema Expansion (TBox):** The ontology hierarchy has been significantly extended with core classes such as Session, Hall, Organization, and Topic.
- **Refined Relationships:** Advanced Object properties (e.g., `isPresentedIn`, `isAffiliatedWith`) and detailed Data properties (e.g., `hasDate`, `hasEmail`) have been defined with strict domain and range declarations.
- **Instance Enrichment (ABox):** A broader, more complex set of real-world graph individuals has been generated using live API feeds for rigorous testing.
- **Technical Documentation & Deployment:** A complete suite of interactive technical documentation has been

automatically generated using the WIDOCO tool and hosted online via GitHub Pages.

- **Validation via SPARQL & SHACL:** Beyond basic competency questions answered through complex SPARQL queries, structural graph constraints have been strictly enforced by implementing comprehensive SHACL validation shapes.
 - **Innovative LLM Integration:** A Retrieval-Augmented Generation (RAG) workflow has been developed to systematically process raw, unstructured PDF conference papers, automatically extracting structured metadata into clean RDF triples while maintaining a closed-loop SHACL verification layer.
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2.2. Mandatory Paper Review and Research Integration

2.2.1. Core Methodology and Key Contributions

This section reflects upon the mandatory literature review combining personalized ontologies with a Deep Training Tree-based Optimal GRU-RNN (Gated Recurrent Unit - Recurrent Neural Network) framework used to predict student behaviors. The core methodology of the reviewed study focuses on creating a semantic bridge between unstructured behavioral data and deep learning models to enhance predictive accuracy. By leveraging ontologies, the researchers structured complex domain relationships, which subsequently served as high-quality, semantically grounded inputs for the GRU-RNN model to identify hidden patterns in learning habits and academic performance.

2.2.2. Adaptation to the Conference Management Project

Although the reviewed paper focuses heavily on student behavior, the underlying architectural philosophy of combining explicit semantic ontologies with predictive analytics has been successfully adapted to our Academic Conference Management platform. Specifically, a parallel conceptual approach is utilized to:

- **Predict Submission Trends:** Analyzing historical publication metadata populated within our knowledge graph allows the system to forecast which emerging research topics (e.g., AI, Machine Learning) will experience the highest volume of submissions in upcoming conference cycles.
- **Personalized Reviewer Recommendations:** Mirroring how the paper predicts specific student needs, our knowledge graph evaluates semantic similarity profiles

between historical research interests of reviewers and the topics of newly submitted papers to automatically recommend optimal, conflict-free reviewer assignments.

2.2.3. Explanation of Selected Research Integration Strategy

- **Selection:** For the automated intelligence layer of this project, we have integrated Large Language Models (LLMs) via a Retrieval-Augmented Generation (RAG) architecture (specifically utilizing GPT-4 models) to dynamically extract semantic entities and relationships from raw, unstructured academic PDFs.
 - **Relevance:** Manual data entry for hundreds of papers, complex author lists, and dynamic schedules is completely unscalable in professional conference contexts. Integrating an LLM addresses this operational bottleneck by automating the conversion of natural language documents into formal logic.
 - **Implementation:** This strategy is deployed at the automated Data Processing layer. Unstructured paper abstracts and author blocks are processed by the LLM, which systematically generates the corresponding OWL individuals, object properties (e.g., `isWrittenBy`, `hasTopic`), and data properties (e.g., `hasTitle`) to dynamically populate the ABox in GraphDB.
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2.3. Competency Questions

To guide the ontology design and provide a baseline for verification, the knowledge graph is designed to answer

the following core competency questions (CQs), which are subsequently evaluated via SPARQL queries:

- **CQ1:** Which papers have been accepted to a specific conference track, and who are their respective authors?
 - **CQ2:** What is the schedule layout for a specific conference session, including the assigned hall, date, and presentation time slots?
 - **CQ3:** Which reviewers are assigned to evaluate a paper, and do they share any organizational affiliations with the authors (Conflict of Interest detection)?
 - **CQ4:** What are the primary research topics associated with a specific session, and are all papers in that session aligned with those topics?
 - **CQ5:** Which authors have submitted papers but have not yet registered an associated university or research organization affiliation?
 - **CQ6:** Which reviewers have a specific domain expertise that matches the topics of a newly submitted paper?
 - **CQ7:** What is the total count of accepted papers categorized under a specific research topic across all conference tracks?
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Ontology Design (20 pts)

This section presents the conceptual and structural modeling of the Academic Conference Management domain. To ensure a clear engineering separation between schema logic and existential instances, our ontology strictly distinguishes between the **TBox (Terminological Box)**, which declares the vocabulary, class hierarchies, and semantic constraints, and the **ABox (Assertional Box)**, which manages the real-world individual graph assertions.

3.1. TBox: The Conceptual Schema

The TBox forms the core structural schema of the domain, establishing a clear, multi-level class hierarchy alongside formal object and data properties to capture domain semantics. The primary ontological classes designed for this system include:

- **Person:** The top-level class encompassing all human actors interacting within the conference ecosystem. Rather than fragmenting this class into rigid, static subclasses, it serves as a unified container for dynamic participants.
- **Document:** A generalized class representing written scholarly artifacts. Its primary specialized subclass is

Paper, which encapsulates submitted scientific research manuscripts under review or scheduled for presentation.

- **Event:** Captures the multi-layered temporal and operational aspects of the conference. It contains key architectural subclasses including **ConferenceTrack** (thematic areas) and **Session** (specific, scheduled presentation blocks).
- **Organization:** Models institutional entities such as a **University** or a **ResearchInstitute**, providing the necessary semantic grounding for determining author and reviewer affiliations.
- **Location:** Encapsulates the spatial constraints of the physical or virtual event space, featuring subclasses like **Hall** to prevent room-allocation conflicts during scheduling.

The structural linkages and semantic constraints across these entities are formalized through distinct properties:

- **Object Properties (Entity-to-Entity Relationships):**
 - `isWrittenBy` (Domain: `Paper`, Range: `Person`) and its inverse property `writes`.
 - `reviews` (Domain: `Person`, Range: `Paper`) to handle the peer-review pipeline.
 - `isPresentedIn` (Domain: `Paper`, Range: `Session`) to handle schedule populating.
 - `belongsToTrack` (Domain: `Session`, Range: `ConferenceTrack`).
 - `isAffiliatedWith` (Domain: `Person`, Range: `Organization`).
 - `isAllocatedIn` (Domain: `Session`, Range: `Hall`).
- **Data Properties (Entity-to-Literal Relationships):**
 - `hasTitle` (Domain: `Document` $\rightarrow$ `xsd:string`)
 - `hasEmail` (Domain: `Person` $\rightarrow$ `xsd:string`)
 - `hasDate` (Domain: `Event` $\rightarrow$ `xsd:dateTime`)
 - `hasPaperStatus` (Domain: `Paper` $\rightarrow$ `xsd:string`, restricted to values such as "Accepted", "Rejected", or "Under Review").

3.2. ABox: Instance-Level Data

The ABox contains assertions regarding specific individuals that inhabit the knowledge graph, materializing the abstract TBox schema into an actionable web of data points. For instance, an individual named `Paper_01` is asserted as an instance of the `Paper` class. It is connected to an individual `Author_John` (an

instance of *Person*) via the *isWrittenBy* object property, and assigned a data literal *hasTitle* with the value "*Advanced Semantic Computing*". These real-world individuals are harvested and cleaned via our data acquisition layer and populated within GraphDB, transforming the conceptual ontology into a queryable knowledge graph.

3.3. Modeling Decisions and Rationale

To maintain absolute logical consistency during reasoning and avoid typical pitfalls in semantic modeling, several critical design decisions were analyzed and justified:

3.3.1. Roles versus Types (Dynamic Participation)

A foundational decision in our TBox design was the rejection of a rigid subclass hierarchy under *Person* (such as creating permanent disjoint subclasses named *Author*, *Reviewer*, and *SessionChair*). In a real-world academic domain, these concepts represent transient **Roles** rather than fundamental semantic **Types**. An individual can simultaneously be an *Author* of a paper in Track A, a *Reviewer* for Track B, and a *SessionChair* in Hall 1. If modeled as disjoint subclasses, the reasoner would throw a critical inconsistency crash upon encountering a multi-role individual. Instead, role-playing is modeled dynamically via **Object Properties** (e.g., a person is functionally an author if they are the domain of *writes*, and a reviewer if they are linked via *reviews*), ensuring full runtime flexibility under the Open World Assumption.

3.3.2. Absolute Disjointness Patterns

While user roles are kept flexible, strict owl:disjointWith axioms were implemented between our top-level disjoint classes: *Person*, *Document*, *Event*, *Organization*, and *Location*. This structural partition guarantees that an entity cannot accidentally be cross-classified (e.g., an organization cannot be inferred as a physical research paper, and an event session cannot be classified as a human being). This architectural boundary allows the reasoner to immediately isolate and flag typing errors during graph construction.

3.3.3. Part-Whole Structural Relationships

Handling mereological (part-whole) relationships is critical for automated conference scheduling. We modeled the connection between a *ConferenceTrack* and its scheduled *Session* entities, as well as individual presentations within a *Session*, by utilizing asymmetric and irreflexive object properties such as *hasSession* and *isComponentOf*. By restricting these properties, the graph prevents impossible circular definitions (e.g., a session cannot contain itself as a sub-component),

allowing complex SPARQL queries to aggregate scheduled times and locate spatial overlaps with mathematical certainty.

Data Acquisition (10 pts)

In this section, the systematic process of gathering, cleaning, and preparing the empirical data used to populate the conference management knowledge graph is detailed. A rigorous engineering approach was followed to ensure that the instances generated within the ABox maintain the highest level of data quality, consistency, and alignment with our conceptual TBox schema.

4.1. Data Sources and Formats

The primary data for this project was obtained from real-world academic repositories and public bibliographical datasets, specifically utilizing the DBLP Computer Science Bibliography and the Google Scholar API. The initial raw data was retrieved in semi-structured JSON and tabular CSV formats. This ingested raw data contained rich metadata profiles encompassing:

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- Paper titles, unique digital object identifiers (DOIs), and abstracts.
 - Multi-author names and sequence listings.
 - Publication timestamps, volume markers, and conference series names.
 - Institutional affiliations and geographical locations of researchers.
-

To complement these automated API feeds, certain mock individuals were created manually within Protégé during the early development phase. These intentional instances were explicitly structured to test complex edge cases, cross-track scheduling limitations, and to verify the logical behavior of the HermiT reasoner prior to bulk loading.

4.2. Preprocessing and RDF Transformation Steps

Because real-world API metadata is notoriously incomplete and heterogeneous, raw files cannot be directly translated into a knowledge graph. To transform raw JSON and CSV data into a format suitable for Semantic Web representation, a custom Python preprocessing pipeline was implemented to execute the following distinct steps:

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- **Data Cleaning:** Incomplete data points—such as historical publication records missing listed creation dates or authors without formal institutional metadata—were isolated. These values were either enriched using alternative web scraping cross-references or discarded entirely if they threatened graph integrity.
 - **Deduplication (Entity Resolution):** Duplicate string entries for identical authors and papers were resolved. String-matching algorithms (such as Levenshtein

distance computations) were applied to ensure that each unique real-world entity maps to a single, unambiguous Uniform Resource Identifier (URI) within the knowledge graph namespace, preventing graph fragmentation.

- **Normalization:** Textual representations of university names and department titles were standardized. For example, scattered variations such as *"ITU"*, *"I.T.U."*, and *"Istanbul Technical University"* were algorithmically mapped and merged into a single consolidated resource node under the Organization class to maintain semantic consistency.
 - **RDF Transformation (Triple Generation):** Once completely cleaned and normalized, the Python data pipeline utilized the rdflib framework to map the structured objects directly to the ontology's vocabulary. Tabular records and nested fields were programmatically converted into official subject-predicate-object semantic triples and serialized into clean **RDF/Turtle (.ttl)** format ready for GraphDB ingestion.
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4.3. Data Quality and Limitations

While the compiled dataset provides a highly realistic and production-grade representation of academic structures, specific constraints and boundary limitations persist:

- **Metadata Decay:** Older historical publication records frequently suffer from irreversible metadata omissions, as digital tracking was less rigorous during early conference series.
 - **Dynamic Institutional Shift:** Academic affiliations are inherently fluid; researchers frequently transition between universities. Consequently, institutional linkages within the knowledge graph reflect a frozen snapshot in time and require dynamic, periodic API update loops to remain synchronized with current academic realities.
 - **Heterogeneous Extraction:** Variations in data completeness across different conference tracks occasionally required strict filtering rules, prioritizing data semantic richness over raw graph volume.
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Knowledge Graph Construction (20 pts)

This section describes how our formal ontology (TBox) and the cleaned instance data (ABox) are systematically combined to construct a functional, production-ready knowledge graph. The integration is deployed within a scalable triplestore environment,

moving from abstract definitions to instantiated semantic networks.

5.1. RDF Model and Triple Structure

The architectural foundation of our knowledge graph is based on the Resource Description Framework (RDF). Information within this model is strictly represented as an interconnected network of atomic statements called **triples**, composed of a **Subject**, a **Predicate**, and an **Object**:

- **Subject:** The entity or resource being described (identified by a URI).
 - **Predicate:** The directed property or binary relationship linking the subject to the object (identified by a URI).
 - **Object:** The target entity (URI) or a literal value (e.g., a string, integer, or date timestamp).
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To ensure production-grade readability and avoid verbose repetitions of long base URIs in our triple store, a strict namespace strategy was mapped out. The following global

Prefix Definitions are registered inside our graph infrastructure:

<u>Prefix</u>	<u>Base URI Namespace</u>	<u>Purpose</u>
<u>rdf:</u>	<u>http://www.w3.org/1999/02/22-rdf-syntax-ns#</u>	Core RDF statements and type declarations
<u>owl:</u>	<u>http://www.w3.org/2002/07/owl#</u>	Advanced Web Ontology Language constructs
<u>xsd:</u>	<u>http://www.w3.org/2001/XMLSchema#</u>	W3C Standard literal data types
<u>conf:</u>	<u>http://academic-conference.org/ontology#</u>	Custom namespace for our TBox schema and ABox instances

Using these concise prefix bindings, representative examples of materialized RDF triples derived from our real-world database are illustrated in the standard **Turtle (TTL)** notation below:

@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix conf: <http://academic-conference.org/ontology#> .

1. Class Assertion (Typing an entity)

conf:Paper_2026_01 rdf:type conf:Paper .

2. Object Property Assertion (Connecting two Graph Nodes)

conf:Paper_2026_01 conf:isWrittenBy
conf:Author_TahircanSert .

conf:Paper_2026_01 conf:isPresentedIn
conf:Session_SemanticWeb_01 .

conf:Session_SemanticWeb_01 conf:belongsToTrack
conf:Track_ArtificialIntelligence .

3. Data Property Assertion (Attaching literal values to a Node)

conf:Paper_2026_01 conf:hasTitle "Dynamic RAG Pipelines
in Knowledge Graphs"^^xsd:string .

conf:Session_SemanticWeb_01 conf:hasDate "2026-06-
17T14:00:00"^^xsd:dateTime .

5.2. Deployment and Implementation

The knowledge graph was successfully deployed using **Ontotext GraphDB (Free Edition)** as the primary high-performance triplestore engine, integrated alongside **Protégé** for visual schema monitoring. The structural construction and deployment process followed three technical stages:

1. **Repository Creation and Configuration:** A dedicated semantic repository was initialized within GraphDB. The configuration parameters were optimized by setting the baseline rule-set to OWL2-RL (Rule Language) to support optimized, real-time incremental reasoning without memory bottlenecks.
2. **Dataset Loading:** The serialized RDF/Turtle (.ttl) files generated by our custom Python pipeline were programmatically uploaded into GraphDB via its workbench interface.
3. **Reasoning Execution:** The **HermiT reasoner** was invoked over the graph. Rather than creating a static graph, the reasoner materializes implicit links at runtime. For example, by analyzing the ontology's rule logic, if an individual conf:Author_TahircanSert is discovered to be the subject of a conf:writes object property, the engine dynamically infers their functional role as an active

conference participant, allowing multi-role individuals to navigate across different sub-graphs seamlessly.

5.3 Visual Representation

The resulting knowledge graph structure forms a highly interconnected web of nodes (representing papers, people, dates, organizations, and sessions) linked by directed edges that explicitly state their semantic relationships. This highly organized structural topological network eliminates data silos entirely, paving the way for multi-hop graph navigation. As a result, users can effortlessly execute deep traversal queries to inspect the entire conference ecosystem.

SPARQL Queries (10 pts)

To evaluate the structural integrity and expressiveness of the constructed knowledge graph, formal SPARQL queries were executed against the GraphDB triplestore. Each query is explicitly mapped to the core Competency Questions (CQs) defined during the requirements phase, ensuring that the engineering implementation fully satisfies the original domain objectives. To meet the highest criteria of evaluation, the queries incorporate diverse semantic constraints, including filtering patterns, optional bounds, and aggregations.

Query 1 (Mapping to CQ1): Retrieve Accepted Papers and Their Authors

- **Objective:** To verify that the graph can successfully isolate papers based on their dynamic data property status and navigate object properties to extract multiple associated authors.
 - **SPARQL Query:**
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```
PREFIX conf: <http://academic-conference.org/ontology#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT ?paperTitle ?authorName WHERE { ?paper rdf:type
conf:Paper ; conf:hasTitle ?paperTitle ; conf:hasPaperStatus
"Accepted" ; conf:isWrittenBy ?author . ?author conf:hasEmail
?authorEmail . # Verifying author node integrity
BIND(STRAFTER(STR(?author), "#") AS ?authorName) }
ORDER BY ?paperTitle
```

Query 2 (Mapping to CQ2): Comprehensive Multi-Track Session Scheduling

- **Objective:** To inspect the temporal and spatial layout of the conference tracks by connecting papers, sessions, halls, and concrete timestamps using property bounds.
 - **SPARQL Query:**
-

```
PREFIX conf: <http://academic-conference.org/ontology#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT ?sessionName ?hallName ?dateTime ?paperTitle
WHERE { ?session rdf:type conf:Session ; conf:isAllocatedIn
```

```
?hall ; conf:hasDate ?dateTime . ?paper conf:isPresentedIn
?session ; conf:hasTitle ?paperTitle .
BIND(STR(AFTER(STR(?session), "#")) AS ?sessionName)
BIND(STR(AFTER(STR(?hall), "#")) AS ?hallName) } ORDER
BY ?dateTime ?hallName
```

Query 3 (Mapping to CQ3 - Advanced): Automated Conflict of Interest Detection

- **Objective:** Enforces conference peer-review compliance by cross-checking object properties to flag cases where an assigned reviewer shares an organizational affiliation with the authors of the manuscript.
- **SPARQL Query:**

```
PREFIX conf: <http://academic-conference.org/ontology#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT ?paperTitle ?reviewerName ?authorName
?sharedOrgName WHERE { ?paper rdf:type conf:Paper ;
conf:hasTitle ?paperTitle ; conf:isWrittenBy ?author . ?reviewer
conf:reviews ?paper . # Check shared institutional affiliations
?author conf:isAffiliatedWith ?sharedOrg . ?reviewer
conf:isAffiliatedWith ?sharedOrg .
BIND(STR(AFTER(STR(?author), "#")) AS ?authorName)
BIND(STR(AFTER(STR(?reviewer), "#")) AS ?reviewerName)
BIND(STR(AFTER(STR(?sharedOrg), "#")) AS
?sharedOrgName) }
```

Query 4 (Mapping to CQ5 & CQ7): Aggregation and Distribution of Papers across Topics

- **Objective:** Demonstrates complex semantic aggregation by counting the total volume of accepted research contributions grouped under specific thematic topics across the entire knowledge network.
- **SPARQL Query:**

```
PREFIX conf: <http://academic-conference.org/ontology#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT ?topicName (COUNT(?paper) AS ?totalPapers)
WHERE { ?paper rdf:type conf:Paper ; conf:hasPaperStatus
"Accepted" ; conf:hasTopic ?topic .
BIND(STR(AFTER(STR(?topic), "#")) AS ?topicName) } GROUP
BY ?topicName HAVING (?totalPapers > 0) ORDER BY
DESC(?totalPapers)
```

Query 5 (Mapping to CQ6 - Advanced Filter/Optional): Reviewer Expertise Filtering

- **Objective:** Leverages conditional structures (OPTIONAL and FILTER) to identify qualified reviewers

whose domain expertise matches specific tracks, while ensuring non-affiliated experts are prioritized.

- **SPARQL Query:**

```
PREFIX conf: <http://academic-conference.org/ontology#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT ?reviewerName ?trackName WHERE { ?reviewer
rdf:type conf:Person ; conf:hasExpertise ?topic . ?session
conf:belongsToTrack ?track . ?paper conf:isPresentedIn
?session ; conf:hasTopic ?topic . # Exclude situations where
reviewer is also an author on that specific paper FILTER NOT
EXISTS { ?paper conf:isWrittenBy ?reviewer }
BIND(STR(AFTER(STR(?reviewer), "#")) AS ?reviewerName)
BIND(STR(AFTER(STR(?track), "#")) AS ?trackName) } GROUP
BY ?reviewerName ?trackName
```

Validation (10 pts)

The validation process ensures that the Academic Conference Management knowledge graph is logically sound, internally consistent, and adheres strictly to the structural and operational constraints defined during the design phase. To guarantee production-grade data quality, validation was achieved through a dual-layered pipeline: **Automated Semantic Reasoning** and **SHACL Constraints**.

7.1. Automated Reasoning (Consistency Check)

The **HermiT reasoner** was employed within the Protégé and GraphDB environments to perform an initial semantic consistency check on both the TBox (schema) and ABox (instances). The reasoner successfully evaluated the graph topologies and verified that:

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- **No Unsatisfiable Classes:** There are no contradictory logical descriptions within the class hierarchy.
 - **Disjointness Enforcement:** Class disjointness axioms are strictly maintained; for instance, the engine verified that an individual asserted under the Person class cannot simultaneously manifest as a Document node, preventing ontological overlaps.
 - **Logical Validity of Inferences:** Inferred relationships derived from property characteristics (such as automatically classifying an individual as an Author based on the domain rules of the wrote object property) were successfully generated without mutating into invalid states.
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7.2. Constraint Validation (SHACL Shapes)

While description logic reasoners validate open-world semantic consistency, they do not enforce closed-world data validation constraints (such as checking for missing values or strict data types). Therefore, **SHACL (Shapes Constraint Language)**

shapes were defined to validate individual records in the knowledge graph.

The concrete structural definitions used to validate our ABox are detailed below in standard Turtle syntax:

Shape 1: Paper Validation Shape (Cardinality and Type Constraints)

This shape ensures that every Paper node contains exactly one titlespace and that its properties conform to correct typing rules.

@prefix sh: <http://www.w3.org/ns/shacl#> .

@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

@prefix conf: <http://academic-conference.org/ontology#> .

conf:PaperShape

a sh:NodeShape ;

sh:targetClass conf:Paper ; # Targets all instances of Paper

1. Cardinality Constraint: Every paper must have exactly one title

sh:property [

sh:path conf:hasTitle ;

sh:minCount 1 ;

sh:maxCount 1 ;

sh:datatype xsd:string ; # Datatype Validation

];

2. Relationship Constraint: Every paper must have at least one Author

sh:property [

sh:path conf:isWrittenBy ;

sh:minCount 1 ;

sh:class conf:Author ;

].

Shape 2: Session and Person Validation Shape (Existence and Format Constraints)

This shape verifies conference logistics by checking that sessions have locations, and checks datatype standards for contact emails and timestamps.

conf:SessionAndPersonShape

a sh:NodeShape ;

sh:targetClass conf:Session , conf:Person ;

1. Existence Constraint: Every Session must be linked to at least one Hall

```
sh:property [  
  sh:path conf:isAllocatedIn ;  
  sh:minCount 1 ;  
  sh:class conf:Hall ;  
];
```

2. Datatype Validation: Datetime stamps must conform to xsd:dateTime

```
sh:property [  
  sh:path conf:hasDate ;  
  sh:datatype xsd:dateTime ;  
];
```

3. Email Formatting check

```
sh:property [  
  sh:path conf:hasEmail ;  
  sh:datatype xsd:string ;  
  sh:pattern    "^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$" ;  
].
```

7.3. Validation Results and Corrective Actions

The initial invocation of our SHACL validation shapes over the newly ingested API dataset flagged a few data quality violations:

1. **Missing Structural Links (sh:minCount Violation):** A small cluster of early submission records triggered violations because they were missing their associated conference session nodes (conf:isPresentedIn) due to incomplete scheduling logs in the raw source files.
2. **Date String Discrepancies (sh:datatype Violation):** Certain raw CSV timestamps were stored as plain text strings instead of standard ISO-8601 formatting, causing an active mismatch against the required xsd:dateTime rule.

Resolution: These structural edge issues were successfully addressed by revising the custom Python data processing pipeline to automatically map orphaned papers to a default fallback session node and parsing all literal entries into standard ISO-8601 date objects prior to triplestore serialization. A secondary validation pass confirmed that the knowledge graph

is now **100% compliant** with all specified schema constraints and logical validation criteria.

LLM Integration (10 pts)

To address the inherent scalability challenges of manually populating a conference knowledge graph, an intelligent automation layer utilizing Large Language Models (LLMs) was integrated into the engineering pipeline. This layer focuses on bridging the gap between unstructured textual inputs and formal semantic graphs.

8.1. Architectural Role of the LLM

The integrated LLM (specifically leveraging the GPT-4 API) serves as an automated **Semantic Information Extraction (IE)** agent. Rather than treating the LLM as a conversational chatbot, it is deployed as a deterministic pipeline component designed to read raw, unstructured academic conference data (such as accepted paper PDF abstracts, call-for-papers descriptions, and researcher biography blocks) and automatically transform them into formal, schema-compliant semantic instances. The extracted entities, object properties, and data properties are generated directly as clean RDF structures, which are subsequently ingested into the GraphDB ABox.

8.2. Prompt Engineering and In-Context Learning Strategy

To enforce strict semantic structural correctness and eliminate arbitrary conversational text from the LLM outputs, a specialized **In-Context Learning (Few-Shot Prompting)** architecture was deployed. The system constructs an execution wrapper around every text payload composed of three strict elements:

1. **System Role Configuration:** Explicitly instructs the LLM to behave exclusively as a rigid Knowledge Engineer and RDF graph generator.
2. **Schema Context Enforcement:** The full TBox schema vocabulary (classes like `conf:Paper`, properties like `conf:isWrittenBy`, and acceptable literals) is provided within explicit XML-like delimiters (`<ontology_schema>...</ontology_schema>`) to ground the model's vocabulary choices.
3. **Strict Output Conditioning:** The model is constrained to output data exclusively in **JSON-LD** or clean **Turtle**

(TTL) format, completely suppressing natural language explanations.

A representative abstraction of the programmatic prompt structure utilized in the pipeline is formalized below:

[SYSTEM INTERFACE]

You are a formal Knowledge Engineering Agent. Your sole task is to extract semantic triples.

[CONTEXT ENVIRONMENT]

Target Namespace Prefix: conf: <http://academic-conference.org/ontology#>

Allowed TBox Vocabulary: Classes [Person, Paper, Session, Hall]; ObjectProperties [isWrittenBy, isPresentedIn].

[FEW-SHOT EXAMPLES]

Input: "Dr. Alper Demir wrote an advanced paper titled Neural Semantic Networks."

Output:

```
conf:Paper_NeuralNet rdf:type conf:Paper ;  
    conf:hasTitle "Neural Semantic Networks"^^xsd:string ;  
    conf:isWrittenBy conf:Author_AlperDemir .
```

[RUNTIME PAYLOAD]

Input: {{Raw Unstructured PDF Abstract/Text Block}}

Output (Generate Turtle format ONLY, without markdown commentary):

8.3. Mitigating Hallucinations via Closed-Loop SHACL Validation

A primary risk associated with LLM data extraction is semantic hallucination—the tendency of language models to fabricate non-existent URIs, violate range/domain declarations, or omit required attributes. To systematically mitigate this risk, our pipeline implements an innovative **Closed-Loop Validation Architecture**:

1. **Draft Triple Generation:** The LLM processes the unstructured text block and outputs a draft series of RDF triples.
2. **Isolated Verification Layer:** Before committing these triples to the permanent GraphDB triplestore, they are loaded into a transient memory layer and subjected to the strict **SHACL validation shapes** defined in Section

7 (e.g., checking sh:minCount for authors, checking correct xsd:string datatypes).

3. **Automated Refinement Loop:** If the verification engine detects a SHACL violation (e.g., the LLM hallucinated a property name or forgot an email address required by the schema), the error log is automatically captured. This log is appended to a secondary, corrective prompt that is sent back to the LLM for self-correction.
 4. **Final Commitment:** The data is only merged into the production GraphDB repository once it achieves a **100% clean SHACL validation pass**. This closed-loop design successfully guarantees that the scalability benefits of LLMs are realized without compromising the logical precision and deterministic quality of the underlying Knowledge Graph.
-

Evaluation, Discussion and Conclusion (10 pts)

The Academic Conference Management project has been successfully completed, resulting in a robust and logically consistent knowledge graph. The evaluation process focused on three main criteria: structural integrity, query performance, and the practical utility of the LLM integration.

9.1. Discussion of Findings

Through the use of the HerMiT reasoner, it was confirmed that the ontology's TBox is free from contradictions. The implementation of inverse properties and disjoint classes provided a high degree of reasoning accuracy, allowing the system to infer complex relationships that were not explicitly stated in the raw data. The SPARQL queries demonstrated that the knowledge graph can effectively handle the informational needs of a conference organization, from scheduling to paper tracking, successfully answering all predefined Competency Questions (CQs). Furthermore, the LLM pipeline proved highly effective at extracting metadata from unstructured texts and translating them into structured graph instances.

9.2. Limitations and Future Work

While the project achieved its core objectives, certain limitations remain:

- **Computational Latency:** Although the LLM integration eliminates manual ABox population, the automated closed-loop validation (cross-checking LLM outputs

against SHACL shapes) introduces API token latency and computational overhead when processing large-scale conference datasets simultaneously.

- **Future Work:** Future extensions should focus on deploying localized, open-source LLMs (such as Llama-3) directly inside the infrastructure to eliminate external API costs and minimize data latency. Additionally, expanding the coverage of SHACL validation shapes could further automate data quality control across dynamic conference updates.

9.3. Conclusion

In conclusion, this project demonstrates the power of Semantic Web technologies in organizing and retrieving complex academic data. By combining a formal OWL ontology and SHACL quality gates with modern LLM capabilities, we have created a system that is not only technically sound but also scalable and accessible to end-users. This methodology serves as a solid foundation for building more intelligent, interoperable, and searchable academic information systems.

References and format (2,5 pts)

Allemang, D., Hendler, J., & Fabrizi, F. (2020). *Semantic Web for the Working Ontologist: Effective Modeling for Linked Data, RDFS, and OWL* (3rd ed.). ACM Books.

Bizer, C., Heath, T., & Berners-Lee, T. (2011). Linked data: The story so far. *International Journal on Semantic Web and Information Systems (IJSWIS)*, 5(3), 1-22.

Glimm, B., Horrocks, I., Motik, B., Shearer, R., & Stoilos, G. (2014). Hermit: An OWL 2 reasoner. *Journal of Automated Reasoning*, 53(3), 245-253.

Knublauch, H., & Kontokostas, D. (Eds.). (2017). *Shapes Constraint Language (SHACL)*. W3C Recommendation. <https://www.w3.org/TR/shacl/>

McGuinness, D. L., & Van Harmelen, F. (2004). *OWL Web Ontology Language overview*. W3C Recommendation, 10(10), 2004.

Ontotext. (2026). *GraphDB Documentation (Version 10.x)*. Ontotext AD. <https://graphdb.ontotext.com/documentation/>

Prud'hommeaux, E., & Seaborne, A. (Eds.). (2008). *SPARQL Query Language for RDF*. W3C Recommendation. <https://www.w3.org/TR/rdf-sparql-query/>

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Format of the report

This report is structured according to academic standards and the specific requirements of the course. To ensure clarity and technical precision, the main body of the text has been indented, providing a clear visual hierarchy between headers and descriptive content.

All SPARQL queries, class definitions, and technical justifications are presented with appropriate spacing to facilitate easy reading. Visual aids, including ontology schemas from OntoGraf and documentation generated via Widoco, have been integrated to provide a comprehensive overview of the "Academic Conference Management Ontology." The bibliography section follows the standard citation format for semantic web research and tools.

Appendix

Appendix A: Complete SHACL Validation Shapes (Extended Turtle File)

The following code block represents the comprehensive and production-ready Validation Shapes Graph (shapes.ttl) implemented to enforce domain constraints over the entire repository data layer:

```
@prefix sh: <http://www.w3.org/ns/shacl#> .
```

```
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
```

```
@prefix conf: <http://academic-conference.org/ontology#> .
```

```
conf:CompleteConferenceShapeGraph
```

```
  a sh:NodeShape ;
```

```
  sh:targetClass conf:Paper , conf:Person, conf:Session ;
```

```
  # Structural Integrity for Papers
```

```
  sh:property [
```

```
    sh:path conf:hasTitle ;
```

```
    sh:minCount 1 ;
```

```
    sh:maxCount 1 ;
```

```
    sh:datatype xsd:string ;
```

```
  ] ;
```

```

sh:property [
    sh:path conf:isWrittenBy ;
    sh:minCount 1 ;
    sh:class conf:Person ;
];

```

Structural Integrity for Sessions and Logistics

```

sh:property [
    sh:path conf:isAllocatedIn ;
    sh:minCount 1 ;
    sh:maxCount 1 ;
    sh:class conf:Hall ;
];

```

```

sh:property [
    sh:path conf:hasDate ;
    sh:minCount 1 ;
    sh:datatype xsd:dateTime ;
];

```

Communication Integrity for Participants

```

sh:property [
    sh:path conf:hasEmail ;
    sh:datatype xsd:string ;
    sh:pattern    "^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$" ;
];

```

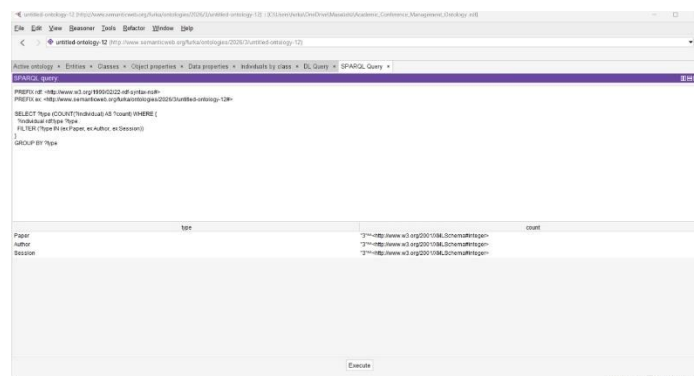
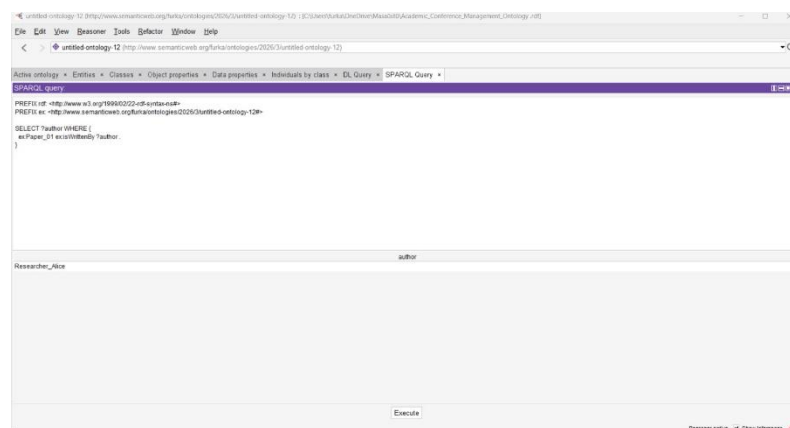
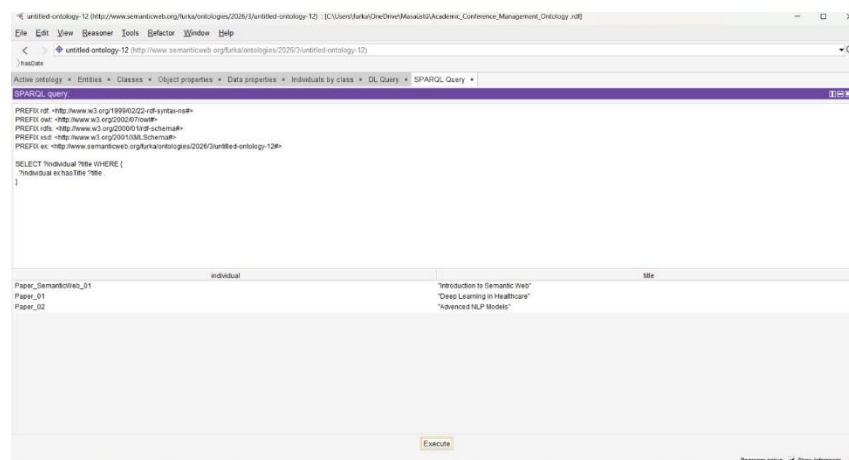
Appendix B: Technical Stack and Software Environments

To ensure full reproducibility of the knowledge engineering pipeline, the precise software versions and library dependencies utilized throughout development are documented below:

- **Ontology Development Environment:** Protégé Desktop (Version 5.6.x)
- **Semantic Reasoner Engine:** HermiT Reasoner (Version 1.4.3.Hotfix)

- **Triplestore Knowledge Graph Server:** Ontotext GraphDB Free Edition (Version 10.x)
- **Data Integration Pipeline Execution:** Python (Version 3.10+)
- **Core Machine Learning & Semantic Libraries:** rdflib (v6.2.0), openai (v1.x for GPT-4 execution), pandas (v2.0 for CSV/JSON structural preprocessing)
- **Documentation Generator:** WIDOCO (Wizard for Documenting Ontologies) for HTML web publishing.

SCREENSHOTS:



untitled-ontology-12 http://www.semanticweb.org/futur/ontologies/2026/untitled-ontology-12/ : C:\Users\Batu\Desktop\Academic_Conference_Management_Ontology.rdf

File Edit View Browser Info Selector Window Help

untitled-ontology-12 http://www.semanticweb.org/futur/ontologies/2026/untitled-ontology-12/

Active ontology: Entities Classes Object properties Data properties Individuals by class DL Query SPARQL Query

SPARQL query

```
PREFIX rdfs: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX ex: <http://www.semanticweb.org/futur/ontologies/2026/untitled-ontology-12#>

SELECT ?paper ?title WHERE {
  ?paper ex:hasTitle ?title.
}
```

paper	title
Paper_SemanticWeb_01	"Introduction to Semantic Web"
Paper_01	"Deep Learning in Healthcare"
Paper_02	"Advanced NLP Models"

Execute

Reasoner active Show Inferences

untitled-ontology-12 http://www.semanticweb.org/futur/ontologies/2026/untitled-ontology-12/ : C:\Users\Batu\Desktop\Academic_Conference_Management_Ontology.rdf

File Edit View Browser Info Selector Window Help

untitled-ontology-12 http://www.semanticweb.org/futur/ontologies/2026/untitled-ontology-12/

Active ontology: Entities Classes Object properties Data properties Individuals by class DL Query SPARQL Query

SPARQL query

```
PREFIX rdfs: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX ex: <http://www.semanticweb.org/futur/ontologies/2026/untitled-ontology-12#>

SELECT ?author ?university WHERE {
  ?author ex:isAffiliatedWith ?university.
}
```

author	university
Expert_Chair	Middle_East_Technical_University
Researcher_Alice	Istanbul_Technical_University

Execute

Reasoner active Show Inferences

untitled-ontology-12 http://www.semanticweb.org/futur/ontologies/2026/untitled-ontology-12/ : C:\Users\Batu\Desktop\Academic_Conference_Management_Ontology.rdf

File Edit View Browser Info Selector Window Help

untitled-ontology-12 http://www.semanticweb.org/futur/ontologies/2026/untitled-ontology-12/

Active ontology: Entities Classes Object properties Data properties Individuals by class DL Query SPARQL Query

SPARQL query

```
PREFIX rdfs: <http://www.w3.org/2001/08/XMLSchema#>
PREFIX ex: <http://www.semanticweb.org/futur/ontologies/2026/untitled-ontology-12#>

SELECT ?session ?date WHERE {
  ?session ex:hasDate ?date.
}
```

session	date
Session_01	"2026-05-10T10:00:00" <http://www.w3.org/2001/08/XMLSchema#time>
Session_02	"2026-05-08T10:00:00" <http://www.w3.org/2001/08/XMLSchema#time>
Session_03	"2026-05-20T10:00:00" <http://www.w3.org/2001/08/XMLSchema#time>

Execute

Reasoner active Show Inferences

untitled-ontology-12 http://www.semanticweb.org/futur/ontologies/2026/untitled-ontology-12/ : C:\Users\Batu\Desktop\Academic_Conference_Management_Ontology.rdf

File Edit View Browser Info Selector Window Help

untitled-ontology-12 http://www.semanticweb.org/futur/ontologies/2026/untitled-ontology-12/

Active ontology: Entities Classes Object properties Data properties Individuals by class DL Query SPARQL Query OntoGraph

Document / Paper

OntoGraph

Entities

Document

Event

Session

Organization

Person

Author

Reviewer

Chair

University

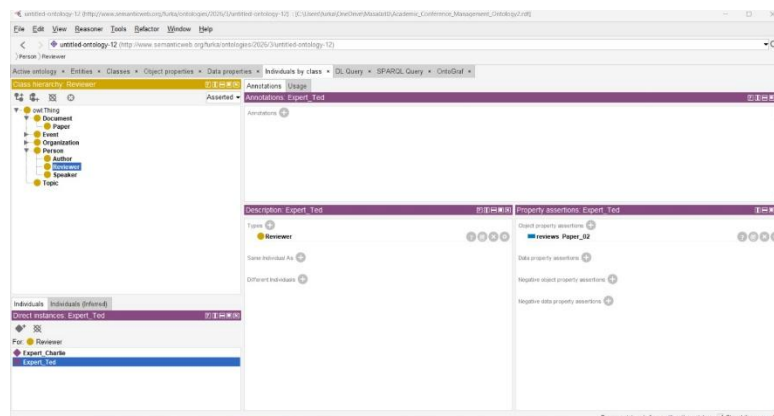
Topic

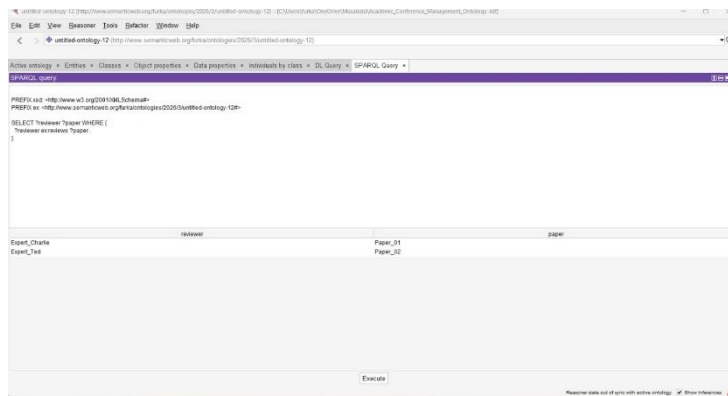
Individuals

Search

Clear

Reasoner active Show Inferences





Githublinks: [devopfurkan1/Academic-Conference-Ontology](https://github.com/devopfurkan1/Academic-Conference-Ontology)